

Particle Filter vs. Extended Kalman Filter Localization for a Mobile Manipulator

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Reliable localization is a critical prerequisite for mobile manipulation, yet algorithms must frequently contend with accumulating odometry drift and perceptual ambiguity in structured environments. This study presents a comparative evaluation of four estimators: dead reckoning, an Extended Kalman Filter (EKF), a Monte Carlo Localization Particle Filter (PF), and a hybrid in which the EKF tracks the robot while a parallel PF detects kidnapping and re-seeds the EKF. These are evaluated on a Clearpath Husky mobile manipulator equipped with a UR5e arm and Robotiq 2F-85 gripper operating in a simulated ROS 2 factory environment. Relying solely on imperfect wheel odometry and 2D LIDAR scans, the filters are evaluated over a three-waypoint route across a sensitivity grid of varying noise injection levels and particle counts. The objective is to quantify the trade-off between the computational efficiency of the EKF during nominal navigation and the robustness of the Particle Filter when recovering from adversarial multimodal scenarios such as global localization and simulated kidnapping.

I. Overview

Serving as the paper’s introduction, this section will introduce the mobile manipulation platform, a Clearpath Husky A200 carrying a UR5e arm and a Robotiq 2F-85 gripper, equipped with a front-mounted 270° 2D LIDAR and wheel odometry. It will provide a high-level summary of the localization problem in the simulated ROS 2/Gazebo open-source factory environment, explaining how the robot must rely on imperfect wheel odometry and noisy LIDAR scans to determine its pose. Finally, it will state the paper’s core objective: a comparative study testing the claim that a Particle Filter can recover in multimodal scenarios where an EKF fails, while the EKF matches PF accuracy at a lower compute cost during benign runs.

Planned Literature Integration:

- **Macenski, S., et al. (2022) [1]:** This paper will be used for **background** on ROS 2, the middleware that connects the simulator, sensors, and estimators in this study.

II. Problem Statement

Providing necessary background on the state estimation problem, this section will formalize the robot’s localization challenges by identifying specific sources of uncertainty. First, it will describe the unbounded error accumulation in wheel odometry caused by wheel slip on the skid-steer base, which comes from skid-steer turning and from Gazebo’s wheel-slip plugin configured with nonzero slip compliance to increase odometry uncertainty, and by injected odometry noise. Second, it will detail the perceptual ambiguity of LIDAR scans (noise-free in Gazebo, so range noise is injected) in a highly structured, repetitive indoor factory environment. Third, it will explain how unknown initial poses (requiring global localization) and unmodeled displacements (kidnapping via Gazebo teleportation) create multimodal posteriors that violate standard single-Gaussian assumptions.

Planned Literature Integration:

- **Thrun, S., Fox, D., Burgard, W., & Dellaert, F. (2001) [2]:** This paper will be used for **background**, as it characterizes the global localization and kidnapped robot problems, the multimodal posteriors they produce, and the difficulty plain MCL has in recovering from them.

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III. Study Approach

Detailing the methodology, this section will outline the mathematical frameworks for the evaluated estimators: dead reckoning, an adapted EKF (fusing odometry with scan-matched pose observations), and a from-scratch Particle Filter (using a likelihood-field LIDAR measurement model). It will then describe a hybrid estimator built from the two: the EKF produces the published pose, while the PF runs alongside on the same odometry and scans purely as a kidnap detector. The hybrid re-seeds the EKF's prior with the PF's mean only when four tests hold on five consecutive PF updates: the two estimates disagree beyond the 99% χ^2 gate on their combined covariance (squared Mahalanobis distance, 3 degrees of freedom); the PF has converged on one hypothesis (position spread under 0.3 m, heading spread under 0.15 rad); the scan's mean per-beam log likelihood is at least 0.3 higher at the PF's mean than at the EKF's; and the scan fits the PF's mean well in absolute terms (mean per-beam log likelihood at least 0). The last two tests stop a collapsed but wrong particle cloud from dragging a correct EKF away, which happened after kidnaps in an early version without the absolute-fit test. Because the particle cloud is overconfident, its covariance is floored at 0.15 m and 0.08 rad before it becomes the EKF's new prior. The section will argue that the hybrid aims to keep the EKF's nominal accuracy and add the PF's recovery from kidnapping, at the cost of running both filters. It will explicitly acknowledge the implementation asymmetry between the adapted EKF and from-scratch PF as a study limitation. It will establish the evaluation protocol: a three-waypoint factory route driven by the simulator's ground-truth pose, so estimator error never alters the trajectory, with all four estimators processing the same recorded sensor streams, and a sensitivity grid over particle count $N \in \{100, 500, 2000, 5000\}$, odometry motion-noise parameters $\alpha_{1..4} = 0.05k$ with $k \in \{0, 1, 2, 4\}$, and LIDAR range noise $\sigma_r \in \{0.01, 0.03, 0.10\}$ m injected into those streams. Finally, it will justify the 20-trial count per configuration: 20 trials pin each reported average to within about half of the typical run-to-run variation (a 95% confidence interval of roughly ± 0.47 standard deviations), and they separate a filter that reliably recovers from one that reliably fails even with a couple of exceptions, since 18 recoveries in 20 trials bound the true recovery rate above 68% while 2 in 20 bound it below 32% (95% Clopper–Pearson intervals).

Planned Literature Integration:

- **Thrun, S., Burgard, W., & Fox, D. (2005) [3]:** This foundational text will be used for **methodology**, providing the canonical mathematical derivations for both the Extended Kalman Filter and Monte Carlo Localization algorithms.
- **Dellaert, F., Fox, D., Burgard, W., & Thrun, S. (1999) [4]:** This paper introduces the core MCL formulation and will be used for **methodology**, as the formulation the from-scratch Particle Filter follows in localizing against the known map.
- **Leonard, J. J., & Durrant-Whyte, H. F. (1991) [5]:** This paper established EKF localization against an a priori map and will be used for **methodology**, as the basis of the EKF baseline estimator.
- **Gordon, N. J., Salmond, D. J., & Smith, A. F. M. (1993) [6]:** This paper introduced the bootstrap filter and will be used for **methodology**, defining the propagate–weight–resample cycle the Particle Filter implements.
- **Arulampalam, M. S., et al. (2002) [7]:** This tutorial provides a comprehensive overview of sequential Monte Carlo methods and will be used for **background**, specifically justifying the selection of the bootstrap/SIR form for the Particle Filter implementation.

IV. Results

This section will present the quantitative findings from the simulated trials using tables and plots to display position RMSE, heading RMSE, waypoint position error, and per-step compute time across the sensitivity grid. It will feature comparative plots showing mean values with 95% confidence intervals to clearly evaluate the primary hypothesis regarding the accuracy-compute trade-off. Additionally, it will visualize the algorithms' recovery times in the adversarial tests (global localization from a uniform prior and mid-run kidnapping), measuring the simulation timesteps required for each filter to return below the 0.25 m position error threshold.

Planned Literature Integration:

- **Gutmann, J.-S., & Fox, D. (2002) [8]:** This experimental comparison of Kalman filter, Markov, and Monte Carlo localization in accuracy, robustness, and recovery from robot displacement will be used for **comparison**, as the prior result against which this study's findings are checked.

A. Preliminary Results

All numbers in this subsection are preliminary. The 20-seed table is being re-run after a fix to the odometry noise injector, and the sensitivity sweep over N , k and σ_r is still running. Results marked “offline synthetic” come from a 2D test map with simulated odometry and LIDAR, not Gazebo.

Core comparison (Gazebo, 20 seeds). Table 1 will report 20 seeded Gazebo trials per row at the nominal operating point: $N = 2000$ particles, injected odometry noise $k = 1$ ($\alpha_{1..4} = 0.05$) and LIDAR range noise $\sigma_r = 0.03$ m, with the RRT* planner driving the three-waypoint route on ground truth. A first 20-seed run is superseded: the odometry noise injector turned in-place spins into spurious jumps of about 0.7 m and 90° , corrupting the inputs of every estimator. The injector has been fixed and the nominal, kidnap and global-localization scenarios are being re-run for all estimators.

Table 1 Core results, Gazebo, 20 seeds per row ($N = 2000$, $k = 1$, $\sigma_r = 0.03$ m). Mean $\pm$ 95% CI; recovery is to within 0.25 m, with its 95% Clopper–Pearson interval. Pending the re-run with the corrected noise injector.

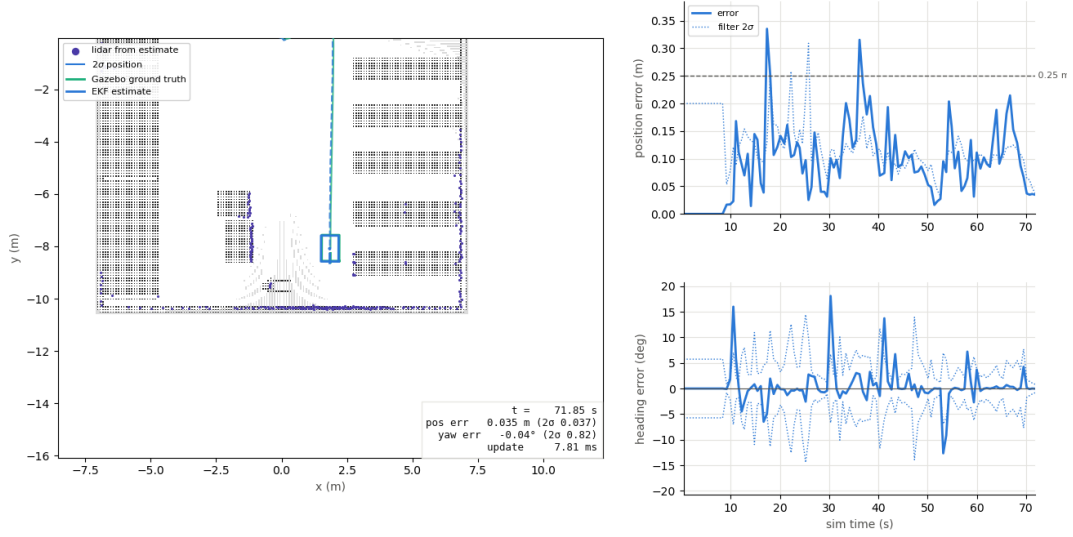
Scenario	Estimator	Pos. RMSE (m)	Head. RMSE ($^\circ$)	Waypoint err. (m)	Update (ms)	Recovered	Recovery (s)
Nominal	EKF			pending re-run			
	PF			pending re-run			
	Dead reckoning			pending re-run			
Kidnap	EKF			pending re-run			
	PF			pending re-run			
Global	EKF			pending re-run			
	PF			pending re-run			
	Dead reckoning			pending re-run			

Test cases (Gazebo, single runs). The two test cases for the operable-code deliverable both pass (Figures 1 and 2), with expected outputs derived independently of the filters: the waypoints in the route’s pose file for the nominal case, and the teleport pose for the kidnap case (Table 2). On the nominal route the EKF finished within 6 cm of every waypoint (0.070 m RMSE) and the PF within 11 cm (0.121 m RMSE), against 3.06 m mean error for dead reckoning. After a teleport at $t = 37$ s the PF was back within 0.25 m in 6.1 s, and the EKF never recovered, finishing 21.5 m from the true pose.

Table 2 Test cases, Gazebo, single runs ($N = 2000$, $k = 1$, $\sigma_r = 0.03$ m). Both pass.

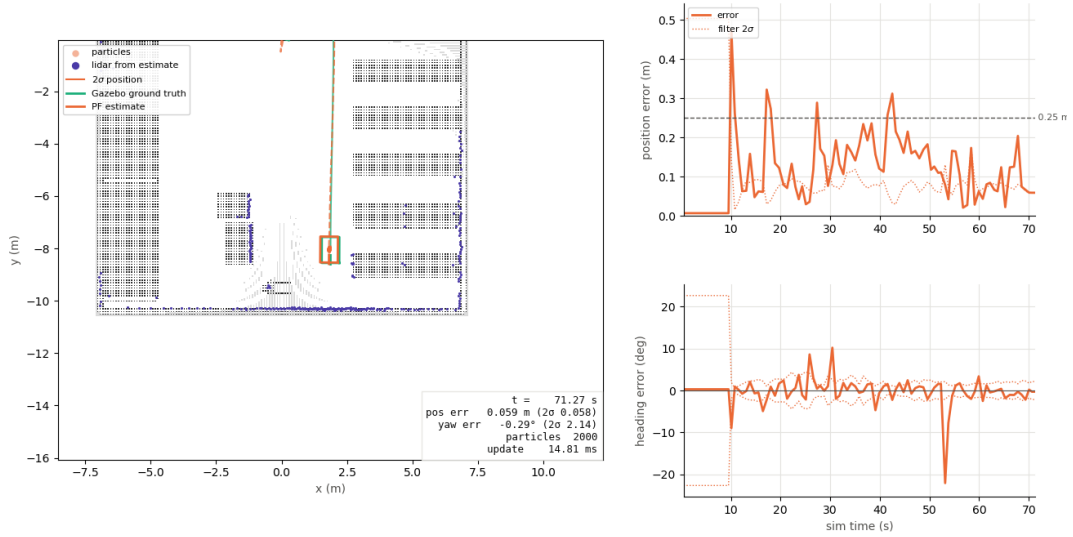
Test case	Estimator	Result	Expected output
1: nominal route	EKF	waypoint errors 0.058 / 0.028 / 0.035 m (RMSE 0.070 m)	route waypoints
	PF	waypoint errors 0.075 / 0.111 / 0.059 m (RMSE 0.121 m)	
	Dead reckoning	3.06 m mean error	
2: kidnap at 37 s	EKF	not recovered, 21.5 m final error	teleport pose
	PF	within 0.25 m after 6.1 s	

EKF vs Gazebo ground truth (nominal, $k=1.0$, $\sigma_r=0.03$ m, seed 0)



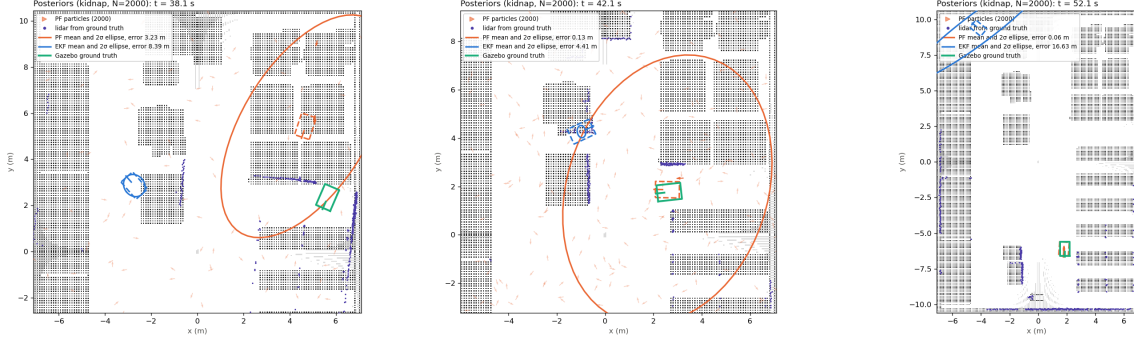
(a) EKF: 0.035 m final error, 2σ bound 0.037 m.

PF vs Gazebo ground truth (nominal, $k=1.0$, $\sigma_r=0.03$ m, seed 0, $N=2000$)



(b) PF ($N = 2000$): 0.059 m final error, 2σ bound 0.058 m.

Fig. 1 Test case 1 (nominal route, Gazebo, seed 0), final frame of each filter's live view. Left of each: occupancy map, ground truth (green), estimate and 2σ ellipse, and the LIDAR scan projected from the estimate. Right: position and heading error against the filter's own 2σ bound. Both filters stay mostly below the 0.25 m threshold, with brief spikes above it, and their errors repeatedly exceed 2σ , the overconfidence discussed below.



(a) $t = 38.1$ s (1 s after): PF 3.23 m, EKF 8.39 m. (b) $t = 42.1$ s (5 s after): PF 0.13 m, EKF 4.41 m. (c) $t = 52.1$ s (15 s after): PF 0.06 m, EKF 16.63 m.

Fig. 2 Test case 2 (kidnap at $t = 37$ s, Gazebo), intermediate posteriors of both filters at the same moments: PF particles and mean with 2σ ellipse (orange), EKF mean and 2σ ellipse (blue), ground truth (green). The injected particles spread over the map find the true pose within seconds, while the EKF’s single Gaussian stays confidently wrong and drifts further away. At 5 s the PF mean is already close although its 2σ ellipse is still several metres wide, because particles remain spread over the map. Live-view recordings of both test cases are written to `results/test_case_1` and `results/test_case_2` by the test-case launch file.

An earlier Gazebo check replayed two recorded 60 s, roughly 21 m drives through both filters without injected noise (PF at $N = 500$). Odometry alone averaged 3.7–4.0 m of position error; the EKF averaged 0.08 m (0.26 m worst, 0.3° heading) at about 4 ms per scan; the PF averaged 0.15–0.18 m (0.38–0.66 m worst, 0.6 – 0.9° heading) at a 1.25 ms median. Both filters were overconfident: the EKF’s position RMSE was about four times its reported σ , and the PF’s about thirteen times. The final paper will report a covariance consistency check (normalized estimation error squared) alongside RMSE so that this is quantified rather than anecdotal.

Offline synthetic tests. Before the Gazebo runs, the evaluation pipeline was exercised on a synthetic 2D map. On the nominal route the EKF reached 0.03 m and the PF 0.09 m position RMSE. After a kidnapping the PF recovered in 11 s and the EKF did not (Figures 3 and 4, which add the error-versus-time traces the Gazebo posterior snapshots lack). Global localization from a uniform prior with 2000 particles converged confidently onto the 180° mirror of the true pose, a symmetry failure that the full study will probe with larger particle counts; it is also a concrete example of the perceptual ambiguity described in the Problem Statement.

EKF vs ground truth (synthetic kidnap, offline test)

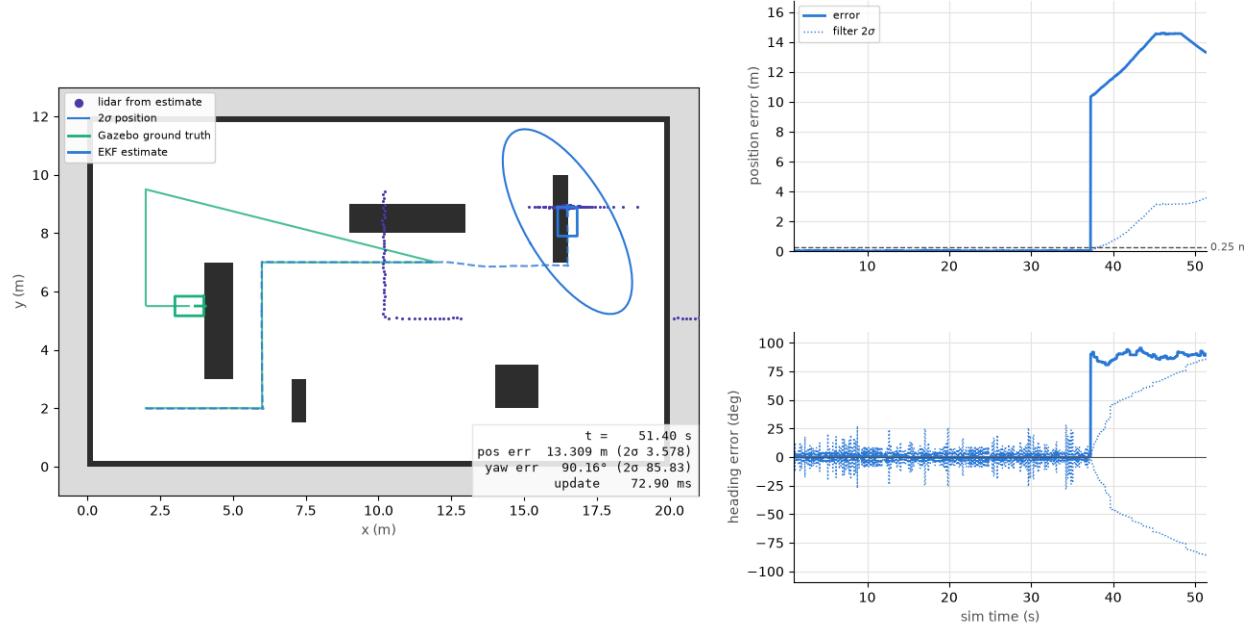


Fig. 3 EKF on the offline synthetic kidnap test. Left: map, ground truth (green), EKF estimate and 2σ ellipse (blue), and the LIDAR scan projected from the estimate. Right: position and heading error against the filter's own 2σ bound. After the teleport at about 37 s the error jumps above 10 m and never returns; the covariance grows but the single Gaussian cannot jump to the true pose.

PF vs ground truth (synthetic kidnap, offline test)

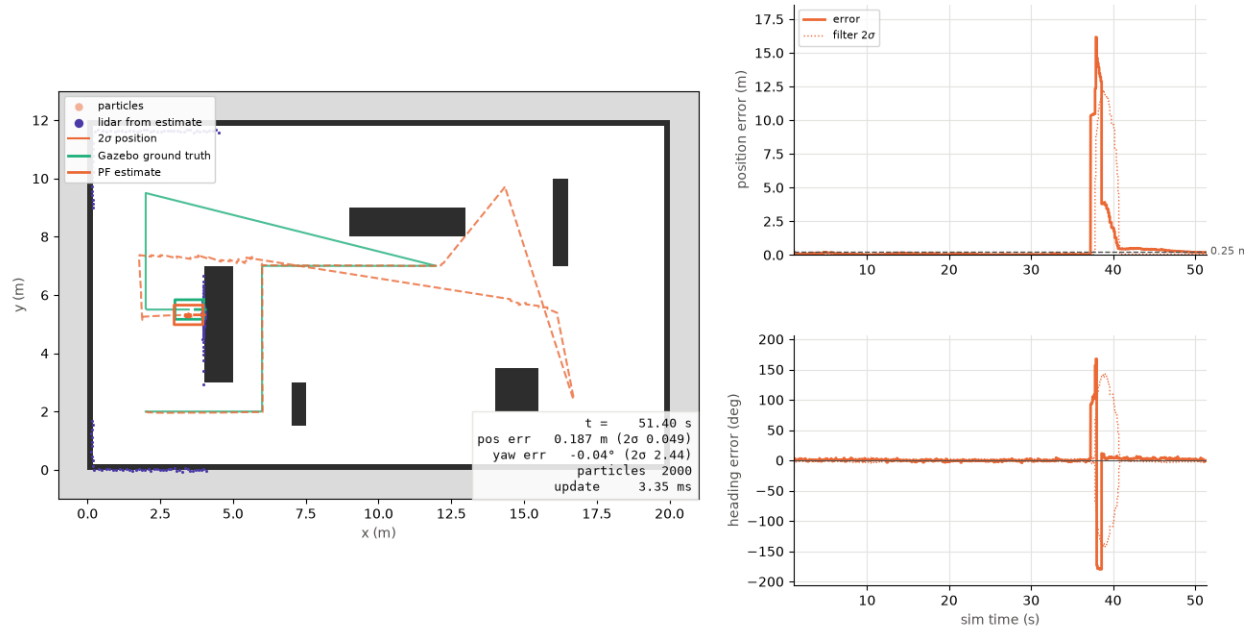


Fig. 4 PF on the same offline synthetic kidnap test ($N = 2000$). The error spikes above 15 m at the teleport and falls back under the 0.25 m threshold within about 11 s as injected particles find the true pose.

Hybrid EKF with PF recovery (offline replay and Gazebo, preliminary). The hybrid was first evaluated offline by replaying one recorded Gazebo drive of the nominal route with 20 noise seeds ($k = 1$, $\sigma_r = 0.03$ m, $N = 500$) and simulating two kidnaps at $t = 50$ s by shifting every filter’s belief: 1.5 m in x and y (k1), and 3 m and -2 m with a 60° rotation (k2). A filter counts as recovered once it stays within 0.25 m for 3 s. Table 3 shows that the hybrid kept the EKF’s tracking accuracy with no re-seeds on any nominal run, and recovered from the kidnaps as often as the PF and sooner. These runs included two later fixes, one to the EKF and one to the PF; without them, the PF lost track on 18 of 20 nominal seeds, and the scan tests vetoed all 269 of its disagreements with the EKF, so the hybrid still made no false re-seeds.

Table 3 Hybrid estimator, offline replay of one recorded Gazebo drive, 20 seeds ($k = 1$, $\sigma_r = 0.03$ m, $N = 500$). Recovery counts with median recovery time. Preliminary.

Estimator	Nominal RMSE (m)	False re-seeds	Kidnap k1 recovered	Kidnap k2 recovered
EKF	0.058	–	3/20 (4.8 s)	4/20 (5.1 s)
PF	0.096	–	16/20 (14.1 s)	19/20 (14.3 s)
Hybrid	0.058	0/20	16/20 (10.5 s)	19/20 (12.4 s)

In Gazebo, test case 2 (teleport kidnap) passed for the hybrid on seeds 0–2, run without wheel slip. The hybrid recovered in 10.6, 6.0 and 7.0 s, against 11.0, 5.1 and 6.6 s for the PF, with exactly one re-seed per run and none before the kidnap; the EKF alone finished 19.9 to 23.3 m from the true pose. The 20-seed Gazebo study will include the hybrid as a fourth estimator.

Still to come. The 20-seed Gazebo results for all four estimators and the one-factor-at-a-time sweep over N , k and σ_r will be added here.

V. Conclusion

The final section will synthesize the experimental results to definitively answer whether the primary hypothesis holds true: that the Particle Filter recovers in both adversarial scenarios where the EKF does not, and that the EKF matches or exceeds nominal-route accuracy at lower per-step compute time. It will then assess whether the hybrid gets the best of both, the EKF’s nominal accuracy and the PF’s recovery, and whether that is worth the cost of running both filters. It will discuss the practical implications of these findings for the robotic system’s downstream waypoint controller and planning pipeline. It will also state the study’s limitations: estimation is planar (3-DOF) on a flat indoor floor; the prior map is built from the same world the robot drives in, so it is known and exact; and that map is a 2D projection of a 3D octomap (obstacles from 0.10 to 5.0 m above the floor) matched against a single LIDAR scan plane, so overhanging structure appears in the map but never in the scan. Lastly, it will propose future work, such as mitigating the noted implementation asymmetry, extending the evaluation to dynamic obstacles within the factory, and a lunar extension with 3D measurement models, an IMU, terrain-dependent wheel slip and a coarse orbital map prior.

Acknowledgments

Anthropic’s Claude was used to review this outline against the course rubric and the instructor’s feedback on the proposal, to draft revisions to it, and to summarize the preliminary trial results and extract figure stills from the filter visualizations. The localization nodes, evaluation tooling and simulation runs reported here were also developed with Claude’s assistance.

References

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